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Password Generator & Auto-Filler
Secure Credential Management System

A USB HID-Based Hardware Password Manager

using STM32 ARM Cortex-M3 Microcontroller

Course: Microcontrollers (PBCST504)

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PROJECT ABSTRACT

1. Objective

To design and implement a plug-and-play USB hardware password manager using the STM32 Blue Pill microcontroller. The system generates high-entropy random passwords, encrypts them locally using AES-128, stores them in internal flash memory, and automatically inputs them into host login fields via USB HID keyboard emulation, protecting credentials from OS-level keyloggers.

2. System Description

This project implements an air-gapped hardware device based on the STM32F103C8T6 (ARM Cortex-M3) development board. Upon pressing a push button connected to GPIO PA0, the firmware executes a pseudo-random number generator (PRNG) seeded with hardware entropy (SysTick timer counter, floating ADC channel noise, and chip Unique Device ID) to generate a 12–16 character password (uppercase, lowercase, numbers, special symbols). The generated password is immediately encrypted using AES-128 and stored in dedicated STM32 internal flash memory. Concurrently, the decrypted password is sent over USB HID keyboard protocol to auto-fill the active credential field on any host PC without requiring host software or drivers.

3. Technical Component Specifications

- **Microcontroller Board:** STM32F103C8T6 Blue Pill (ARM Cortex-M3, 72 MHz, 64 KB Flash, 20 KB SRAM)
- **Programming Interface:** ST-Link V2 In-Circuit Debugger/Programmer over 4-pin SWD interface
- **Hardware Peripherals:** PA0 Push Button (trigger), 10 kΩ pull-down resistor, USB Micro-B data cable, breadboard
- **Software Toolchain:** Arduino IDE with STM32 Core (`STM32duino`), ST-Link Drivers, Embedded C/C++
- **Cryptography Protocol:** AES-128 encryption (NIST FIPS 197 compliant), USB 2.0 Full-Speed HID Keyboard Class

4. Key Features Security Benefits

- **High-Entropy Generation:** Enforces strong 12–16 character mixed passwords, ending reliance on weak or reused passwords.
- **Hardware Anti-Keylogger Protection:** Transmits scan codes directly via USB HID endpoints, insulating credentials from host malware and clipboard monitors.
- **AES-128 Flash Security:** Protects data at rest in internal flash memory against unauthorized physical extraction.

- **Universal Plug-and-Play:** Operates driverless across Windows, macOS, Linux, and Android.

5. Conclusion

The project delivers a secure, low-cost hardware token combining embedded systems programming, cryptographic storage, and USB protocol emulation to solve critical password security vulnerabilities.

Keywords: STM32 Blue Pill, USB HID, AES-128 Encryption, Embedded Security, Hardware Password Manager, Anti-Keylogger.